TIC 178331798

Eclipsing binary candidate dossier - Star Vetter

CLEAN

```
BLS EPHEMERIS
  Period        1.820518 d
  Epoch (t0)    1492.0037 BTJD
  Duration      0.1200 d
  Depth         2.452 %  (24,522 ppm)
  SNR           79.9

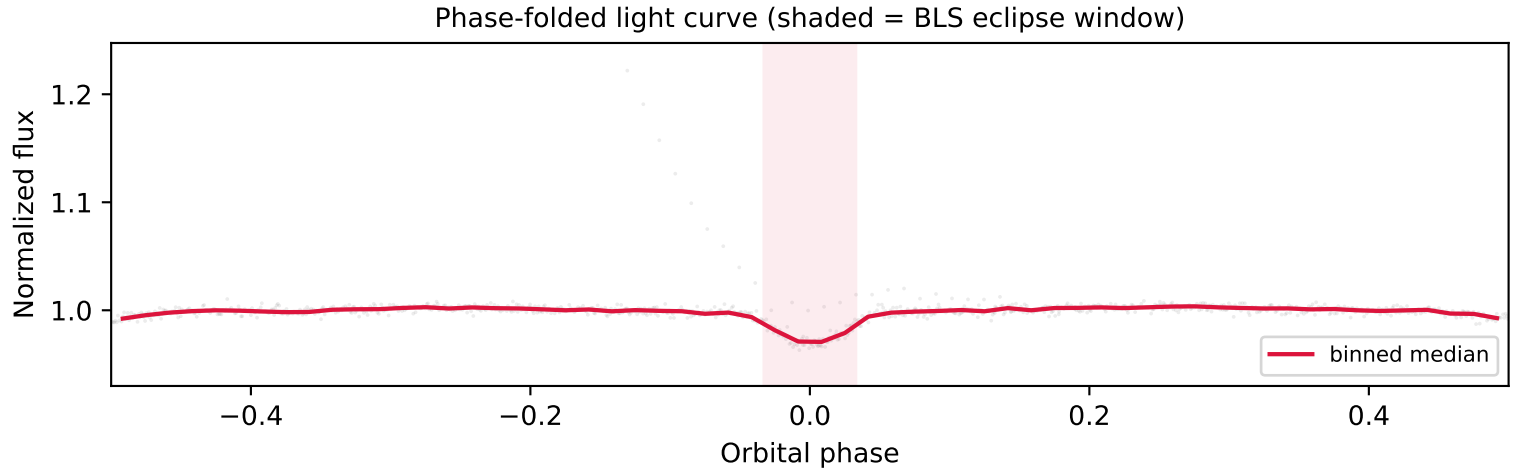
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed      14  (gate >= 3)
  Robust depth ratio     1.01  (gate >= 0.50)
  Robust depth sigma     49.7  (gate >= 7.0)
  Odd/even z-score       0.06  (consistent)
  Secondary eclipse      not detected
```

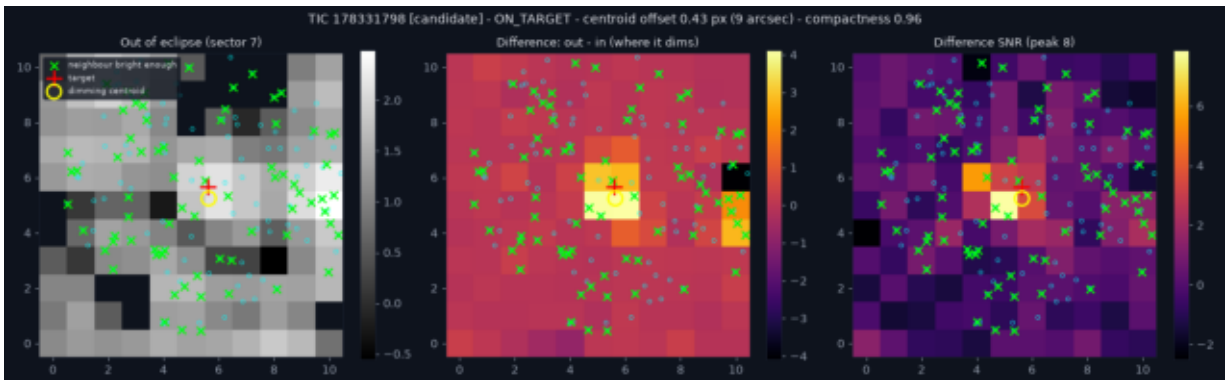
```
CATALOG CROSS-MATCH (novelty)

  VSX                                  no match
  ASAS-SN                              no match
  Gaia eclipsing-binary table          no match
  Gaia variable classification         no match
  TESS EB catalog (Prsa+2022)          no match
  ExoFOP TOI list                      no match

  Verdict: NOVEL
```

```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 7    light-curve dip in this sector: 64.4 sigma    t0 refined by -0.0025 phase
  Difference-image peak SNR   7.86   (gate >= 5.0)      Compactness   0.96   (gate >= 0.50)
  Dimming centroid offset     0.43 px (9 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 178331800
    0.66 px away, dTmag +1.10 - farther from the dimming than the target
```





No caveats - clean detection on every check.